HSC Mathematics Extension 1: Sequences and Series

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"Sometimes it is the people no one imagines anything of who do the things that no one can imagine."

Alan Turing

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1 Introduction

1.1 Project Overview

This booklet is a focused companion for NSW HSC Mathematics Extension 1 students studying sequences and series. It is designed to connect core sequence skills to exam-style problem solving with clear methods and structured solutions.

1.2 Target Audience

This resource is written for Year 12 students in Extension 1, and for tutors and teachers who want a compact booklet with consistent notation and worked examples.

1.3 How to Use This Booklet

- Read the fundamentals review first, then attempt the detailed problems in Part 1.
- Use Part 2 for timed practice and fluency, using hints only when needed.
- Keep a formula sheet while you work, but always justify each step from first principles.
- Track common slips, especially sign errors in geometric sums and indexing mistakes in sigma notation.

1.4 Topics Covered

This booklet develops the following Extension 1 ideas:

- defining sequences by term formula, recursion, and context
- arithmetic sequences and arithmetic series
- geometric sequences and geometric series
- finite sum formulas and sigma notation
- limiting sum of a geometric series when $|r| < 1$
- links between recurring decimals and geometric series
- mixed AP/GP problem solving in HSC-style questions

1.5 Where This Leads Next

Sequence fluency supports later work in calculus, proof, and Extension 2 style reasoning. In particular, convergence arguments and recursive definitions form a bridge to harder topics such as iterative methods and infinite processes.

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2 Fundamentals Review

This section is a compact reference for notation, formulas, and common reasoning patterns used throughout the booklet.

2.1 How Sequences Are Specified

A sequence is an ordered list of numbers, usually written as $(a_n)_{n\geq 1}$. Common ways to define a sequence are:

- explicit formula: a_n written directly in terms of n
- recursive formula: each term written using previous terms
- contextual rule: terms generated from a worded process

Examples.

$$a_n = 3n - 1, \quad b_1 = 2,\ b_{n+1} = b_n + 5, \quad c_n = (-1)^n.$$

2.2 Arithmetic Sequences (AP)

An arithmetic sequence has constant difference d:

$$a_{n+1} - a_n = d.$$

If the first term is a, then

$$a_n = a + (n-1)d.$$

Problem 2.1: Recognizing an AP

Given $a_1 = 7$ and $a_{n+1} = a_n - 3$, find a_{10}.

Solution 2.1

Here $d = -3$, so

$$a_{10} = 7 + (10-1)(-3) = 7 - 27 = -20.$$

2.3 Arithmetic Series

The sum of the first n terms of an AP is:

$$S_n = \frac{n}{2}\big(2a + (n-1)d\big) = \frac{n}{2}(a + l),$$

where l is the nth term.

Problem 2.2: AP Sum

Find $\sum_{k=1}^{20}(5 + 2(k-1))$.

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Solution 2.2

This is an AP with $a = 5$, $d = 2$, and $n = 20$. The 20th term is

$$l = 5 + 19 \cdot 2 = 43.$$

So

$$S_{20} = \frac{20}{2}(5 + 43) = 10 \cdot 48 = 480.$$

2.4 Geometric Sequences (GP)

A geometric sequence has constant ratio r:

$$\frac{a_{n+1}}{a_n} = r \quad (a_n \neq 0).$$

If the first term is a, then

$$a_n = ar^{n-1}.$$

Problem 2.3: Recognizing a GP

If $a_1 = 81$ and $r = \frac{1}{3}$, find a_5.

Solution 2.3

$$a_5 = 81\left(\frac{1}{3}\right)^4 = 81 \cdot \frac{1}{81} = 1.$$

2.5 Geometric Series

For $r \neq 1$, the sum of the first n terms is

$$S_n = a\frac{1 - r^n}{1 - r} = a\frac{r^n - 1}{r - 1}.$$

Use the form that keeps signs clean for your r value.

Problem 2.4: Finite GP Sum

Find $\sum_{k=0}^{6} 3 \cdot 2^k$.

Solution 2.4

This is $a = 3$, $r = 2$, $n = 7$ terms (from $k = 0$ to 6):

$$S_7 = 3\frac{2^7 - 1}{2 - 1} = 3(128 - 1) = 381.$$

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2.6 Limiting Sum of a Geometric Series

If $|r| < 1$, then $r^n \to 0$ as $n \to \infty$, and the infinite sum exists:

$$S_\infty = \frac{a}{1-r}.$$

If $|r| \geq 1$, the geometric series does not converge to a finite limiting sum.

Problem 2.5: Infinite GP

Evaluate $2 + \frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \cdots$.

Solution 2.5

Here $a = 2$, $r = \frac{1}{3}$, and $|r| < 1$, so

$$S_\infty = \frac{2}{1-\frac{1}{3}} = \frac{2}{\frac{2}{3}} = 3.$$

2.7 Recurring Decimals and Geometric Series

Recurring decimals can be written as geometric series. For example,

$$0.\overline{7} = 0.7 + 0.07 + 0.007 + \cdots$$

is geometric with $a = 0.7$ and $r = 0.1$. So

$$0.\overline{7} = \frac{0.7}{1-0.1} = \frac{0.7}{0.9} = \frac{7}{9}.$$

2.8 Sigma Notation Quick Rules

$$\sum_{k=1}^{n}(u_k + v_k) = \sum_{k=1}^{n} u_k + \sum_{k=1}^{n} v_k, \quad \sum_{k=1}^{n} cu_k = c\sum_{k=1}^{n} u_k.$$

Index shifts must preserve both the expression and bounds consistently.

Takeaways 2.1

- Always identify whether the question asks for a term a_n, a finite sum S_n, or a limit.
- AP questions depend on constant difference; GP questions depend on constant ratio.
- For infinite geometric sums, check $|r| < 1$ before using $S_\infty = \frac{a}{1-r}$.
- Track indexing carefully in sigma notation to avoid off-by-one errors.

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3 Part 1: Problems and Solutions (Detailed)

Part 1 contains full solutions with method explanations.

3.1 Basic Sequence Problems

Problem 3.1: Arithmetic nth term

An arithmetic sequence has first term 4 and common difference 5. Find a_{30}.

Solution 3.1

Use $a_n = a + (n-1)d$:

$$a_{30} = 4 + 29 \cdot 5 = 149.$$

Problem 3.2: Find the common ratio

In a geometric sequence, $a_2 = 12$ and $a_5 = 324$. Find the common ratio r and the first term a_1.

Solution 3.2

From $a_n = ar^{n-1}$,

$$a_2 = ar = 12, \qquad a_5 = ar^4 = 324.$$

Divide to remove a:

$$\frac{a_5}{a_2} = r^3 = \frac{324}{12} = 27 \implies r = 3.$$

Then $a_1 = a = \frac{12}{3} = 4$.

Problem 3.3: Sum of an AP

Find the sum of the first 40 terms of the arithmetic sequence

$$9,\ 13,\ 17,\ \ldots$$

Solution 3.3

Here $a = 9$, $d = 4$, $n = 40$. The 40th term is

$$l = 9 + 39 \cdot 4 = 165.$$

So

$$S_{40} = \frac{40}{2}(9 + 165) = 20 \cdot 174 = 3480.$$

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Problem 3.4: Recurring decimal as a fraction

Express $0.\overline{23}$ as a rational number.

Solution 3.4

$$0.\overline{23} = \frac{23}{99}.$$

A geometric view is

$$0.23 + 0.0023 + 0.000023 + \cdots$$

with first term $\frac{23}{100}$ and ratio $\frac{1}{100}$.

3.2 Medium Sequence Problems

Problem 3.5: Mixed AP constraints

An arithmetic sequence has $a_3 = 11$ and $a_{12} = 47$. Find a_1 and d, then determine S_{25}.

Solution 3.5

Use $a_n = a + (n-1)d$:

$$a + 2d = 11, \qquad a + 11d = 47.$$

Subtract:

$$9d = 36 \implies d = 4,$$

then $a = 11 - 8 = 3$. Now

$$a_{25} = 3 + 24 \cdot 4 = 99,$$

so

$$S_{25} = \frac{25}{2}(3 + 99) = \frac{25}{2} \cdot 102 = 1275.$$

Problem 3.6: Finite GP with unknown index

For the geometric sequence with first term 5 and ratio 2, find n such that

$$S_n = 2555.$$

Solution 3.6

$$S_n = 5\frac{2^n - 1}{2 - 1} = 5(2^n - 1) = 2555.$$

So

$$2^n - 1 = 511 \implies 2^n = 512 = 2^9.$$

Hence $n = 9$.

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3.3 Advanced Sequence Problems

Problem 3.7: Recursive sequence and limiting value

Let (x_n) be defined by

$$x_1 = 16, \qquad x_{n+1} = 2 + \frac{1}{x_n} \quad (n \geq 1).$$

(i) Find exact values of x_2, x_3, x_4.

(ii) Assuming $x_n \to L > 0$, show that $L = 1 + \sqrt{2}$.

(iii) Prove that

$$|x_{n+1} - L| = \frac{|x_n - L|}{Lx_n},$$

and deduce that for $n \geq 2$,

$$|x_{n+1} - L| < \frac{1}{4}|x_n - L|.$$

(iv) Hence show $x_n \to 1 + \sqrt{2}$.

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Solution 3.7

(i)

$$x_2 = 2 + \frac{1}{16} = \frac{33}{16}, \quad x_3 = 2 + \frac{16}{33} = \frac{82}{33}, \quad x_4 = 2 + \frac{33}{82} = \frac{197}{82}.$$

For $n \geq 2$, we have $x_n > 2$ because each term is $2 + \frac{1}{x_{n-1}}$ with positive second part.

(ii) If $x_n \to L$, then taking limits in the recurrence gives

$$L = 2 + \frac{1}{L} \implies L^2 - 2L - 1 = 0.$$

So $L = 1 \pm \sqrt{2}$. Since terms are positive and > 2 from $n \geq 2$, we take

$$L = 1 + \sqrt{2}.$$

(iii)

$$x_{n+1} - L = \left(2 + \frac{1}{x_n}\right) - \left(2 + \frac{1}{L}\right) = \frac{L - x_n}{Lx_n},$$

thus

$$|x_{n+1} - L| = \frac{|x_n - L|}{Lx_n}.$$

Now $L = 1 + \sqrt{2} > 2$ and $x_n > 2$ for $n \geq 2$, so $Lx_n > 4$. Hence

$$|x_{n+1} - L| < \frac{1}{4}|x_n - L|, \quad n \geq 2.$$

(iv) Repeatedly applying the inequality gives geometric decay of the error:

$$|x_n - L| \leq \left(\frac{1}{4}\right)^{n-2} |x_2 - L| \to 0.$$

Therefore $x_n \to L = 1 + \sqrt{2}$.

Takeaways 3.1

- For recursive limits, first solve the fixed-point equation.
- A contraction inequality gives a rigorous convergence proof.
- The same method works for many rational recursions.

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4 Part 2: Problems with Hints and Solutions (Concise)

Part 2 gives extra practice with upside-down hints and concise solutions.

4.1 Basic Sequence Problems

Problem 4.1: Quick AP term

Find a_{18} for the arithmetic sequence $2, 7, 12, \ldots$.

Hint: Use $a_n = a + (n-1)d$ with $a = 2$ and $d = 5$.

Solution 4.1

$$a_{18} = 2 + 17 \cdot 5 = 87.$$

Problem 4.2: Quick GP sum

Find

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{128}.$$

Hint: This is a finite GP with $a = 1$, $r = \frac{1}{2}$, and 8 terms.

Solution 4.2

$$S_8 = \frac{1 - (\frac{1}{2})^8}{1 - \frac{1}{2}} = 2\left(1 - \frac{1}{256}\right) = \frac{255}{128}.$$

4.2 Medium Sequence Problems

Problem 4.3: Indexing with sigma notation

Given

$$\sum_{k=1}^{n} (3k - 1) = 119,$$

find n.

Hint: Split the sum into $3\sum k - \sum 1$.

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Solution 4.3

$$\sum_{k=1}^{n}(3k-1) = 3 \cdot \frac{n(n+1)}{2} - n = \frac{3n^2+n}{2}.$$

Set equal to 119:

$$\frac{3n^2+n}{2} = 119 \implies 3n^2+n-238 = 0.$$

$$\Delta = 1 + 2856 = 2857.$$

This is not a perfect square, so there is no integer n with exact value 119.

Problem 4.4: Infinite geometric interpretation

Evaluate

$$0.4\overline{2} = 0.42222\ldots$$

as a fraction.

Hint: Write as $0.4 + 0.02222\ldots$ and convert the recurring part using a GP.

Solution 4.4

$$0.02222\ldots = \frac{2}{90} = \frac{1}{45}.$$

So

$$0.4\overline{2} = \frac{2}{5} + \frac{1}{45} = \frac{18}{45} + \frac{1}{45} = \frac{19}{45}.$$

4.3 Advanced Sequence Problems

Problem 4.5: Convergence condition

Consider the geometric series

$$\sum_{n=0}^{\infty} 7\left(\frac{m-1}{3}\right)^n.$$

Find all real values of m for which the series converges, and then state its sum.

Hint: Use the condition $|r| < 1$ for convergence of an infinite GP.

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Solution 4.5

Here

$$r = \frac{m-1}{3}.$$

Convergence requires

$$\left|\frac{m-1}{3}\right| < 1 \implies |m-1| < 3 \implies -2 < m < 4.$$

For these values,

$$S_\infty = \frac{7}{1 - \frac{m-1}{3}} = \frac{21}{4-m}.$$

Problem 4.6: AP and GP crossover

A sequence is both arithmetic and geometric. Given $a_1 = 6$ and $a_3 = 6$, find all possible sequences.

Hint: Let AP difference be d and GP ratio be r. Use both models on a_3.

Solution 4.6

AP gives

$$a_3 = a_1 + 2d = 6 + 2d = 6 \implies d = 0,$$

so AP is constant: $a_n = 6$. For GP,

$$a_3 = a_1 r^2 = 6r^2 = 6 \implies r^2 = 1 \implies r = \pm 1.$$

If $r = -1$, terms alternate $6, -6, 6, \ldots$ and are not arithmetic. Hence the only sequence that is both AP and GP is

$$a_n = 6 \quad \text{for all } n,$$

with $d = 0$ and $r = 1$.

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5 Conclusion

Strong sequence work in Extension 1 is not just memorizing formulas. It is about choosing the right model, controlling notation, and checking whether each step is logically justified.

The key habits are:

- define terms clearly, including first term and common difference or ratio
- check indexing before and after every sigma manipulation
- identify whether a question is about terms, partial sums, or limits
- test answers for reasonableness by checking growth and sign behavior

With steady practice, sequences become one of the most reliable scoring areas in Extension 1.

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6 Appendices

6.1 Appendix A: Formula Sheet

- Arithmetic nth term:
$$a_n = a + (n-1)d.$$
- Arithmetic sum of first n terms:
$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l).$$
- Geometric nth term:
$$a_n = ar^{n-1}.$$
- Geometric sum of first n terms ($r \neq 1$):
$$S_n = a\frac{1-r^n}{1-r} = a\frac{r^n - 1}{r-1}.$$
- Infinite geometric sum ($|r| < 1$):
$$S_\infty = \frac{a}{1-r}.$$
- Useful sigma results:
$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}.$$

6.2 Appendix B: Common Sequence Traps

- Mixing up term formulas and sum formulas: a_n and S_n answer different questions.
- Losing track of index starts: check whether the sequence starts at $n = 0$ or $n = 1$.
- Using $S_\infty = \frac{a}{1-r}$ when $|r| \geq 1$: this formula only applies for $|r| < 1$.
- Sign errors in GP sums: keep the exact formula form and substitute carefully.
- Assuming recursion converges without proof: identify monotonicity or contraction before claiming limits.
- Treating recurring decimals as finite truncations: they are infinite geometric expansions.

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